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Preliminary Concepts.

Given:

$A = \{n \in \mathbb{N} : n \text{ is even}\}$

$B = \{n \in \mathbb{N} : n \text{ is prime}\}$

$C = \{n \in \mathbb{N} : n \text{ is a multiple of } 5\}$

1.1 Describe sets

a) $A \cap B$

→ An element must be both even and prime.

The only even prime is 2

$$A \cap B = \{2\}$$

b) $B \cap C$

→ An element must be prime and multiple of 5. The only number is 5.

$$B \cap C = \{5\}$$

c) $A \cup B$

→ All natural numbers that are either prime or even.

$$A \cup B = \{2, 3, 4, 5, 6, 7, 8, 9, 10, \dots\}$$

d) $A \cap (B \cup C)$

→ This means even number that are also either prime or multiple of 5.

even prime, 2
even multiple of 5: 10, 20, 30

1.2 Prove

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Take any x

If $x \in A \cup (B \cap C)$ then either,

$x \in A$, so, $x \in A \cup B$ and $x \in A \cup C$, hence

$x \in (A \cup B) \cap (A \cup C)$ or,

$x \in B$ and $x \in C$

hence, $x \in (A \cup B) \cap (A \cup C)$

Conversely, if $x \in (A \cup B) \cap (A \cup C)$ then,

$x \in A$ or $x \in B$ and $x \in C$, we get

$x \in A$ or $x \in B \cap C$

Hence, $x \in A \cup (B \cap C)$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Take any x

If $x \in A \cup (B \cap C)$ then, $x \in A$ and

$x \in B \cap C$

so either, $x \in B$ or $x \in C$

If $x \in B$, then $x \in A \cup B$

If $x \in C$, then $x \in A \cup C$

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$$

Hence,

Conversely, if $x \in (A \cup B) \cap (A \cup C)$ then either,

$x \in A$ so, $x \in A \cup B$ and $x \in A \cup C$

hence, $x \in A \cup (B \cap C)$

Thus, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

1.3 Define function $f: \mathbb{N} \rightarrow \mathbb{N}$

as one to one but not onto

$$f(n) = 2n$$

This is injective, but not surjective.

onto but not one to one

$$f(n) = \left\lfloor \frac{n}{2} \right\rfloor$$

$$f(1) = 1$$

$$f(2) = 1$$

$$f(3) = 1$$

$$f(4) = 2$$

2. Matrix Operation

a) 3×3 with ones on diagonal

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b) 3×4 with one diagonal

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

c) 4×3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

2.2. Compute AD & DA

$$AD = \begin{bmatrix} 1 \times 2 + 1 \times 0 + 1 \times 0 & 1 \times 0 + 1 \times 3 + 1 \times 0 & 1 \times 0 + 1 \times 0 + 1 \times 4 \\ 1 \times 2 + 1 \times 0 + 1 \times 0 & 2 \times 0 + 2 \times 3 + 2 \times 0 & 3 \times 0 + 3 \times 0 + 3 \times 4 \\ 1 \times 2 + 1 \times 0 + 1 \times 0 & 1 \times 0 + 1 \times 3 + 3 \times 0 & 4 \times 0 + 4 \times 0 + 4 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}$$

Also,

$$DA = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

$$\therefore AD \neq DA$$

2.3. Verify $AB = AC$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 \times 4 + 2 \times 0 & 1 \times 3 + 2 \times 2 \\ 2 \times 4 + 4 \times 0 & 2 \times 3 + 4 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$\therefore AB = AC$$

2.4 Prove $(AB)^T = B^T A^T$

Let $A = (a_{ij})$ be $m \times n$, $B = (b_{ij})$ be $n \times p$

Then, (i, j) = entry of AB is

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

So, the (i, j) entry of $(AB)^T$ is

$$[(AB)^T]_{ij} = (AB)_{ji} = \sum_{k=1}^n a_{jk} b_{ki}$$

Now, consider $B^T A^T$. Its (i, j) entry is

$$[B^T A^T]_{ij} = \sum_{k=1}^n (B^T)_{ik} (A^T)_{kj}$$

$$[B^T]_{ik} = \sum_{l=1}^n b_{li} \quad [A^T]_{kj} = \sum_{l=1}^m a_{lj}$$

So, scalar multiplication is commutative

$$\sum_{k=1}^n b_{li} a_{kj} = \sum_{k=1}^n a_{kj} b_{li}$$

$$[(AB)^T]_{ij} = [B^T A^T]_{ij}$$

$$(AB)^T = B^T A^T$$

2.5 Compute $A^T A$ for

$$A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

Then,

$$A^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

So,

$$A^T A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$= \begin{bmatrix} a^2 + b^2 + c^2 & ad + be + cf \\ a d + b e + c f & d^2 + e^2 + f^2 \end{bmatrix}$$

2.6 Find inverse

$$A = \begin{bmatrix} 5 & 6 \\ u & 5 \end{bmatrix}$$

$$\det(A) = 5 \cdot 5 - 6 \cdot u$$

$$= 25 - 6u$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 5 & -6 \\ -u & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -6 \\ -u & 5 \end{bmatrix}$$

h) $A = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$

$\det(A) = 18 - 12$
 $= 6$

$A^{-1} = \frac{1}{6} \begin{bmatrix} 3 & 4 \\ 3 & 3 \end{bmatrix}$

$= \begin{bmatrix} 1/2 & 2/3 \\ -1/2 & 1 \end{bmatrix}$

27. $A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}, A^{-1} = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$

$AA^{-1} = \begin{bmatrix} 4(1) + 1(-3) & -4 + 4 \\ 3(1) + 1(-3) & 1 - 1 + 4 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$AA^{-1} = \begin{bmatrix} 1 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$
 $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

28.

verify the two 3×3 matrices are inverse

$M = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & 4 \end{bmatrix}, N = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -4 & 1 \\ 3 & 5 & 1 \end{bmatrix}$

$MN = I$
 $NM = I$

where I is the 3×3 identity matrix

$MN = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 4 & 1 \\ 2 & -4 & 1 \\ 3 & 5 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= I$

$NM = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & 4 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

2. I

$$\therefore m^T = 2$$

$$n^T = m$$

3-1 Two variables

a)
$$\begin{cases} x + 2y = 4 \\ 3x - 5y = 1 \end{cases}$$

augmented form

$$\begin{bmatrix} 1 & 2 & | & 4 \\ 3 & -5 & | & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & | & 4 \\ 0 & -11 & | & -11 \end{bmatrix}$$

Now, Divide

$$R_2 \rightarrow R_2 \times (-1)$$

$$\begin{bmatrix} 1 & 2 & | & 4 \\ 0 & 1 & | & 1 \end{bmatrix}$$

Now

$$R_1 \rightarrow R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & | & 2 \\ 0 & 1 & | & 1 \end{bmatrix}$$

$$(x, y) = (2, 1)$$

b)

$$\begin{cases} 5x + y = 13 \\ 3x + 2y = 5 \end{cases}$$

augmented form

$$\begin{bmatrix} 5 & 1 & | & 13 \\ 3 & 2 & | & 5 \end{bmatrix}$$

$$R_2 \rightarrow 5R_2 - 3R_1$$

$$\begin{bmatrix} 5 & 1 & | & 13 \\ 0 & 7 & | & -14 \end{bmatrix}$$

$$R_2 \rightarrow R_2 \div 7$$

$$\begin{bmatrix} 5 & 1 & | & 13 \\ 0 & 1 & | & -2 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 5 & 0 & | & 15 \\ 0 & 1 & | & -2 \end{bmatrix}$$

$$R_1 \rightarrow R_1 \div 5$$

$$\begin{bmatrix} 1 & 0 & | & 3 \\ 0 & 1 & | & -2 \end{bmatrix}$$

$$\therefore (x, y) = (3, -2)$$

9) $\begin{cases} 3x + 2y = -2 \\ x + 4y = 6 \end{cases}$

augmented form

$$\left[\begin{array}{cc|c} 3 & 2 & -2 \\ 1 & 4 & 6 \end{array} \right]$$

swap rows to make convenient

$$\left[\begin{array}{cc|c} 1 & 4 & 6 \\ 3 & 2 & -2 \end{array} \right]$$

now,

$$R_2 \rightarrow R_2 - 3R_1$$

$$\left[\begin{array}{cc|c} 1 & 4 & 6 \\ 0 & -10 & -20 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 4R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 2 \end{array} \right]$$

$$\therefore (x, y) = (-2, 2)$$

10)

$$\begin{cases} 2x + 3y = 6 \\ x - y = 1/2 \end{cases}$$

augmented form

$$\left[\begin{array}{cc|c} 2 & 3 & 6 \\ 1 & -1 & 1/2 \end{array} \right]$$

swap rows

$$\left[\begin{array}{cc|c} 1 & -1 & 1/2 \\ 2 & 3 & 6 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\left[\begin{array}{cc|c} 1 & -1 & 1/2 \\ 0 & 1 & 1 \end{array} \right]$$

So,

$$x = 1/2 \quad y = 1$$

11)

$$\begin{cases} x + 3y = 4 \\ x - 3y = -1 \end{cases}$$

augmented form

$$\left[\begin{array}{cc|c} 1 & 3 & 4 \\ 1 & -3 & -1 \end{array} \right]$$

3.2 3x3 matrix

$$a) \begin{cases} x + y - z = 2 \\ x + z = 7 \\ 2x + y + z = 13 \end{cases}$$

augmented form

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 0 & 1 & 7 \\ 2 & 1 & 1 & 13 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & -1 & 3 & 5 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

make pivot in R_2

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -2 & -5 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

$$R_1 \rightarrow R_1 + R_2, \quad R_2 \rightarrow R_2 + 2R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$\therefore x = 3$$

$$y = 3$$

$$z = 2$$

4. Descriptive

1. Problem :- find $f'(u)$

$$f(u) = 2 + yz + 1(u^2)$$

$$f'(u) = 0 - 1 - \frac{2}{u^3}$$

$$= - \left(\frac{1}{u^2} + \frac{2}{u^3} \right)$$

$$= - \frac{(u+2)}{u^3}$$

$$f(u) = \sqrt{u} \cos u$$

$$= n^{1/2} \cos n$$

$$= \frac{1}{n^2} \cos n$$

$$= \frac{1}{2\sqrt{n}} \cos n - \sqrt{n} \sin n$$

$$f(n) = n^3 e^n$$

$$f'(n) = 3n^2 e^n + n^3 e^n$$

$$= n^2 e^n (n+3)$$

$$f(n) = (3n^2 + 2n + 1)^{-5}$$

Let

$$u = 3n^2 + 2n + 1$$

$$f(n) = u^{-5}$$

$$\frac{d}{dn} (u^{-5}) = 5u^{-4}$$

$$\frac{dy}{dn} = \frac{d}{dn} (3n^2 + 2n + 1)$$

$$= 6n + 2$$

$$\frac{dy}{dn} = \frac{dy}{du} \cdot \frac{du}{dn}$$

$$f'(n) = 5u^{-4} (6n + 2)$$

$$= 5(3n^2 + 2n + 1)^{-4} (6n + 2)$$

$$f(n) = \sin n^n$$

Identity inner and outer function

$$u^v$$

$$u = \sin n$$

$$\frac{d}{dn} (u^v) = v u^{v-1} \cdot \frac{du}{dn}$$

$$\frac{d}{dn} (\sin n) = \cos n$$

$$\frac{d}{dn} \frac{dy}{du} = \frac{dy}{du} \cdot \frac{du}{dn}$$

$$= n (\sin n)^{n-1} (\cos n)$$

$$f(n) = e^{3n^2 + 2n}$$

$$u = (3n^2 + 2n)$$

$$\frac{d}{dn} (e^u) = e^u \frac{du}{dn}$$

$$\frac{dy}{dn} = 6n + 2$$

$$f'(u) = e^{3u^2 + 2u} (6u + 2)$$

$$f(u) = \ln(5u^2 + 3u)$$

$$\frac{du}{du} = \frac{u'}{u}$$

$$u = 5u^2 + 3u$$

$$u' = 10u + 3$$

$$f'(u) = \frac{10u + 3}{5u^2 + 3u}$$

1. Minima and Maxima

$$a) f(u) = u^2 - 4u + 5 \text{ on } [1, 4]$$

$$f'(u) = 2u - 4$$

critical points,

Set derivative to 0

$$2u - 4 = 0$$

$$2u = 4$$

$$u = 2$$

Since $2 \in [1, 4]$ is critical

we check,
 $u = 1, 2, 4$

$$f(1) = 1 - 4 + 5 = 2$$

$$f(2) = 4 - 8 + 5 = 1$$

$$= 1$$

$$f(4) = 16 - 16 + 5 = 5$$

compare values,

$$f(1) = 2$$

$$f(2) = 1$$

$$f(4) = 5$$

Smallest value is 1
largest value is 5

Minimum is 1 at $u = 2$
Maximum is 5 at $u = 4$

$$x \quad \frac{u^2}{2} - 2u \text{ at interval } [-2, 3]$$

$$h'(u) = \frac{u^2}{2} - 2u$$

$$= \frac{u^2 - 4u}{2} = \frac{u(u - 4)}{2}$$

$$\therefore n = 2$$

Since

$2 \in (-2, 3]$ is a critical point

We check,

$$n = -2, 2, 3$$

$$h(-2) = \frac{(-2)^3}{2} - 2(-2)$$

$$= \frac{-8}{2} + 4$$

$$= 0$$

$$h(2) = 2 - 4$$

$$= -2$$

$$h(3) = \frac{3^3}{2} - (2(3))$$

$$= \frac{-3}{2}$$

$$\therefore \text{minimum} = -2 \text{ at } n = 2$$

$$\text{maximum} = 0 \text{ at } n = -2$$

c) $3n^3 - 9n^2 + 7$ at interval $[-1, 2]$

$$f'(n) = 9n^2 - 18n + 0$$

$$= 9n(n-2)$$

$$n = 0 \text{ or } n = 2$$

Both are in interval $[-1, 2]$

We check, $[-1, 0, 2]$

$$f(-1) = 3(-1)^3 - 9(-1)^2 + 7$$

$$= -3 - 9 + 7$$

$$= -5$$

$$f(0) = 3 \cdot 0^3 - 9 \cdot 0^2 + 7$$

$$= 7$$

$$f(2) = 3 \cdot 2^3 - 9 \cdot 2^2 + 7$$

$$= -5$$

$$\therefore \text{minimum} = -5 \text{ at } n = -1 \text{ \& } 2$$

$$\text{maximum} = 7 \text{ at } n = 0$$

d) $f(x) = x^2 - 4x + 6$ on $[0, 3]$

$$f'(x) = 2x - 4 + 0$$

$$= 2x - 4$$

$$2x - 4 = 0$$

$$2(x - 2) = 0$$

$$x = 2$$

we evaluate $f(0, 2, 3)$

$$f(0) = 0^2 - 4(0) + 6$$

$$= 6$$

$$f(2) = 2^2 - 4(2) + 6$$

$$= 2$$

$$f(3) = 3^2 - 4(3) + 6$$

$$= 3$$

∴ minimum $= 2$ at $x = 2$
maximum $= 6$ at $x = 0$

1. Evaluate at all gradients

$$f(x, y) = x^2 + y^2$$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial f}{\partial y} = 2y$$

$$\nabla f = (2x, 2y)$$

$$\text{At } (0, 0) = 2 \cdot 0, 2 \cdot 0 = (0, 0)$$

$$\text{At } (1, 3) = 0 - 1, 3 \cdot 2 = (-1, 6)$$

$$\text{At } (-1, -5) = (-2, -10)$$

2. $f(x, y, z) = x^3 - 2xy^2z + 5xz$

$\nabla = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$

$\frac{\partial f}{\partial x} = 3x^2z - 2y^2z + 5z$

$\frac{\partial f}{\partial y} = -4xy^2z$

$\frac{\partial f}{\partial z} = x^3 - 2xy^2 + 5x$

$\frac{\partial f}{\partial x} = 3x^2z - 2y^2z + 5z$

$\frac{\partial f}{\partial y} = -4xy^2z$

2. derivative

$\frac{\partial}{\partial x} (x^3z) = 3x^2z$

$\frac{\partial}{\partial y} (-2xy^2z) = -4xyz$

$\frac{\partial}{\partial z} (5xz) = 5x$

$\frac{\partial f}{\partial z} = x^3 - 2xy^2 + 5x$

$= (3x^2z - 2y^2z + 5z, -4xyz, x^3 - 2xy^2 + 5x)$

points at (1, 1, -1)

$3(1)^2 - 2(-1)^2 = 1 - 2 = -1$

$-4(1)(1)(-1) = 4$

$1^3 + 5 = 1 + 5 = 6$

$\nabla f(1, 1, -1) = (4, 4, 6)$

At (0, 1, 0)

$3(0)^2 - 2(1)^2 = -2$

$-4(0)(1)(0) = 0$

$0^3 + 5 = 5$

$\nabla f(0, 1, 0) = (-2, 0, 5)$